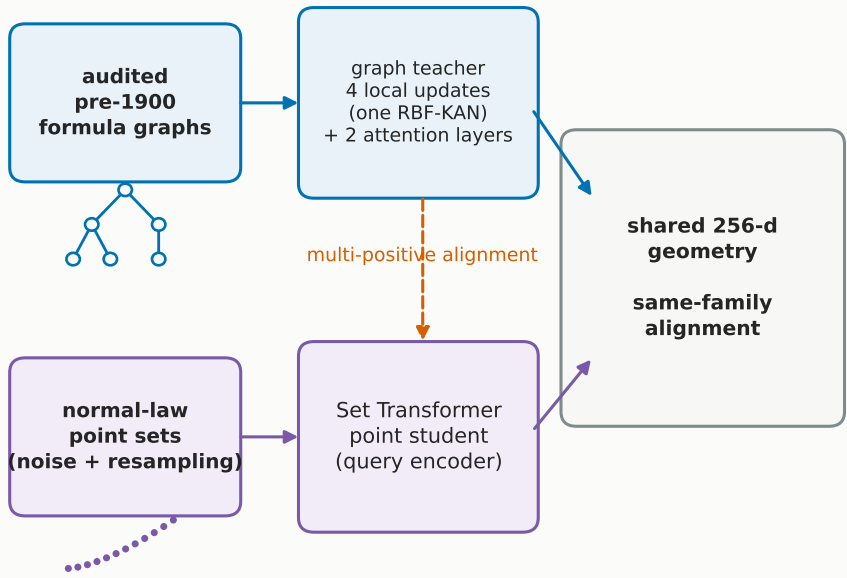


Learning the geometry of an explicit knowledge horizon

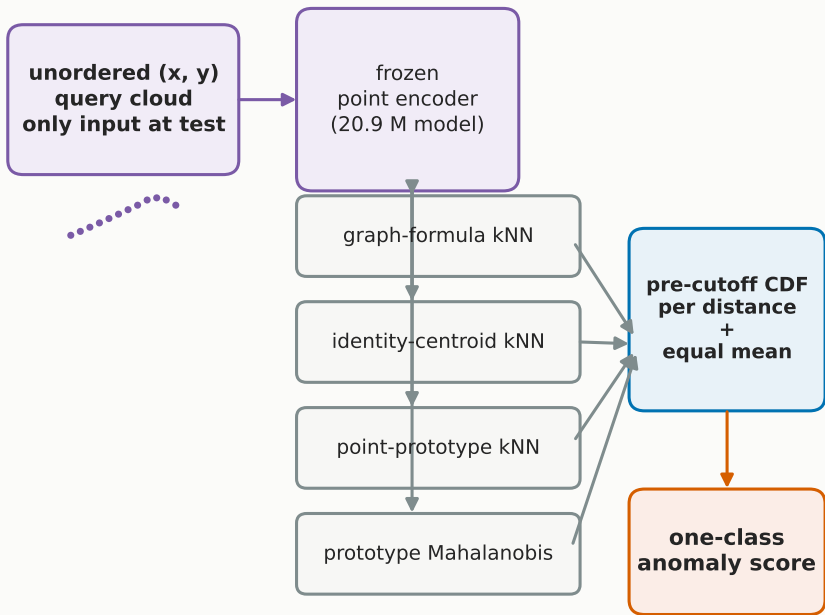
Formula structure supervises representation learning only inside the admitted era; future queries are scored one-class, without a breakdown template.

a Positive-only historical representation learning



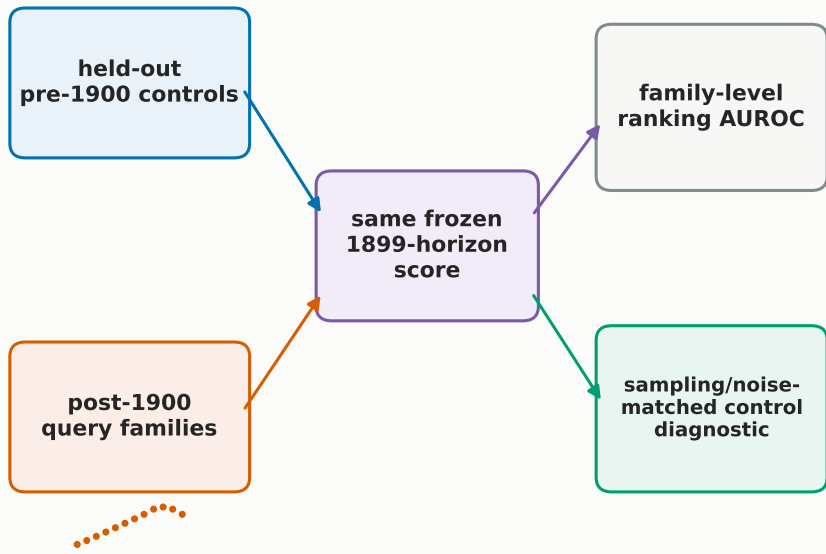
No post-cutoff formulas • no saturation/step/kink labels • no units or dates

b Frozen one-class audit



cloud → formula → source-law family; primary endpoint: grouped AUROC

c Frozen evaluation at the 1899 horizon



Primary: later-family ranking; diagnostic: controls regenerated under the later sampling process